

4.1. Body:  $(0,0,1), (1,0,1), (0,1,1), (1,1,2), (1,-1,0), (-1,1,1)$

$$ax + by + c = z$$

Dosadíme body do rovnice a prevedieme do maticového zápisu

$$(0,0,1) \rightarrow 0a + 0b + c = 1z$$

$$(1,0,1) \rightarrow 1a + 0b + c = 1z$$

$$(0,1,1) \rightarrow 0a + 1b + c = 1z$$

$$(1,1,2) \rightarrow 1a + 1b + c = 2z$$

$$(1,-1,0) \rightarrow 1a - 1b + c = 0z$$

$$(-1,1,1) \rightarrow -1a + 1b + c = 1z$$

$$A = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix} \quad b = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}$$

Maticový zápis  $Au = z$ , kde  $u = (a, b, c)^T$ . To skrípt poznáme  
vetať pre najlepšiu aproximáciu v zmysle najmenších štvorcov  $A^T A u = A^T z$

$$A^T A = \begin{pmatrix} \langle s_1, s_1 \rangle & \langle s_1, s_2 \rangle & \langle s_1, s_3 \rangle \\ \langle s_2, s_1 \rangle & \langle s_2, s_2 \rangle & \langle s_2, s_3 \rangle \\ \langle s_3, s_1 \rangle & \langle s_3, s_2 \rangle & \langle s_3, s_3 \rangle \end{pmatrix} = \begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix}$$

$$\langle s_1, s_1 \rangle = 0^2 + 1^2 + 0^2 + 1^2 + 1^2 + (-1)^2 = 4$$

$$\langle s_1, s_2 \rangle = 0 + 0 + 0 + 1 - 1 - 1 = -1 = \langle s_2, s_1 \rangle$$

$$\langle s_1, s_3 \rangle = 0 + 1 + 0 + 1 + 1 - 1 = 2 = \langle s_3, s_1 \rangle$$

$$\langle s_2, s_2 \rangle = 0^2 + 0^2 + 1^2 + 1^2 + (-1)^2 + 1^2 = 4$$

$$\langle s_3, s_3 \rangle = 1^2 + 1^2 + 1^2 + 1^2 + 1^2 + 1^2 = 6$$

$$\langle s_2, s_3 \rangle = 0 + 0 + 1 + 1 - 1 + 1 = 2 = \langle s_3, s_2 \rangle$$

→ kde prvky matice sú skalárne  
súčin sčítan  
 $s_1, s_2, s_3$

$$A^T z = \begin{pmatrix} \langle s_1, z \rangle \\ \langle s_2, z \rangle \\ \langle s_3, z \rangle \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

$$(A^T A | A^T z) = \left( \begin{array}{ccc|c} 4 & -1 & 2 & 2 \\ -1 & 4 & 2 & 4 \\ 2 & 2 & 6 & 6 \end{array} \right) \sim \left( \begin{array}{ccc|c} 4 & -1 & 2 & 2 \\ -1 & 4 & 2 & 4 \\ 1 & 1 & 3 & 3 \end{array} \right)$$

$$\langle s_1, z \rangle = 0 + 1 + 0 + 2 + 0 - 1 = 2$$

$$\langle s_2, z \rangle = 0 + 0 + 1 + 2 + 0 + 1 = 4$$

$$\langle s_3, z \rangle = 1 + 1 + 1 + 2 + 0 + 1 = 6$$

$$\sim \left( \begin{array}{ccc|c} 1 & 1 & 3 & 3 \\ -1 & 4 & 2 & 4 \\ 4 & -1 & 2 & 2 \end{array} \right) \sim \left( \begin{array}{ccc|c} 1 & 1 & 3 & 3 \\ 0 & 5 & 5 & 7 \\ 0 & -5 & -10 & -10 \end{array} \right) \sim \left( \begin{array}{ccc|c} 1 & 1 & 3 & 3 \\ 0 & 5 & 5 & 7 \\ 0 & 0 & -5 & -3 \end{array} \right) \sim$$

$$\left( \begin{array}{ccc|c} 1 & 1 & 3 & 3 \\ 0 & 5 & 0 & 4 \\ 0 & 0 & -5 & -3 \end{array} \right) \quad c = \frac{3}{5}$$

$$b = \frac{4}{5}$$

$$a = \frac{2}{5}$$

Parametry hľadanej roviny sú  $a = \frac{2}{5}, b = \frac{4}{5}, c = \frac{3}{5}$

$$z = \frac{2}{5}x + \frac{4}{5}y + \frac{3}{5}$$